

JaeSung Han

I have over six years of experience developing Windows applications and device integration software. At Medical IP, I developed C++/Qt medical-imaging client features and delivered the first DeepCatch X release. My work also covers UI test automation, batch processing, and DICOM/PACS reliability.

6+ years in software development · Windows applications · Medical imaging / device integration

Selected projects

01 DeepCatch X Client Development and Product Improvement

Delivered the first release in June 2024 and implemented institution-specific requirements for output, saving, and upload behavior.

02 C++/Qt UI AutoTest Development

Identified dozens of crashes through extended repeat testing and shared daily emails with test results and failure locations.

03 MEDIP Batch Analysis and Export Macro

Allowed processing to continue after individual file failures and recorded per-file status and error messages in JSONL to identify inputs requiring reprocessing.

04 DICOM/PACS Stabilization and C-ECHO Addition

Made server connectivity visible in the product UI and resolved configuration differences and communication errors using reproducible conditions and logs.

Experience

Medical IP | Application / Assistant Manager · May 2023 - Present

Developed medical-imaging client features, UI and batch automation, DICOM/PACS integration, and deployment validation for hospitals and screening sites.

IMPEACH | R&D / Senior Staff · Oct 2019 - Feb 2023

Developed Python/Qt-based test applications and AMX-based integrated control programs. Implemented TCP/IP and Serial-based command/response flows for equipment and external software and handled WATCHOUT integration, connection states, and communication exceptions.

Tech Stack

C/C++ / Qt / MVVM / Python · TCP/IP / Serial / Command/Response / DICOM · AI Result Integration / DICOM Dataset / X-ray / Segmentation · PostgreSQL / Qt SQL / File I/O / GitLab

Contact: Please use the contact details provided in my resume.

DeepCatch X Client Development and Product Improvement

Apr 2024–Jun 2026 · Product Development & Operation

Transitioned TiSepX features and resources into DeepCatch X, implementing image/report output and analysis automation on an inherited MVVM foundation.

Outcome Delivered the first release in June 2024 and implemented institution-specific requirements for output, saving, and upload behavior.

Inherited MVVM foundation

Analyzed the existing screens, state, and data flow on top of the initial View and ViewModel structure.

WW/WL-based imaging output

Applied WW/WL settings to X-ray images and composited AI segmentation masks and annotations into screen and saved result images.

Report output

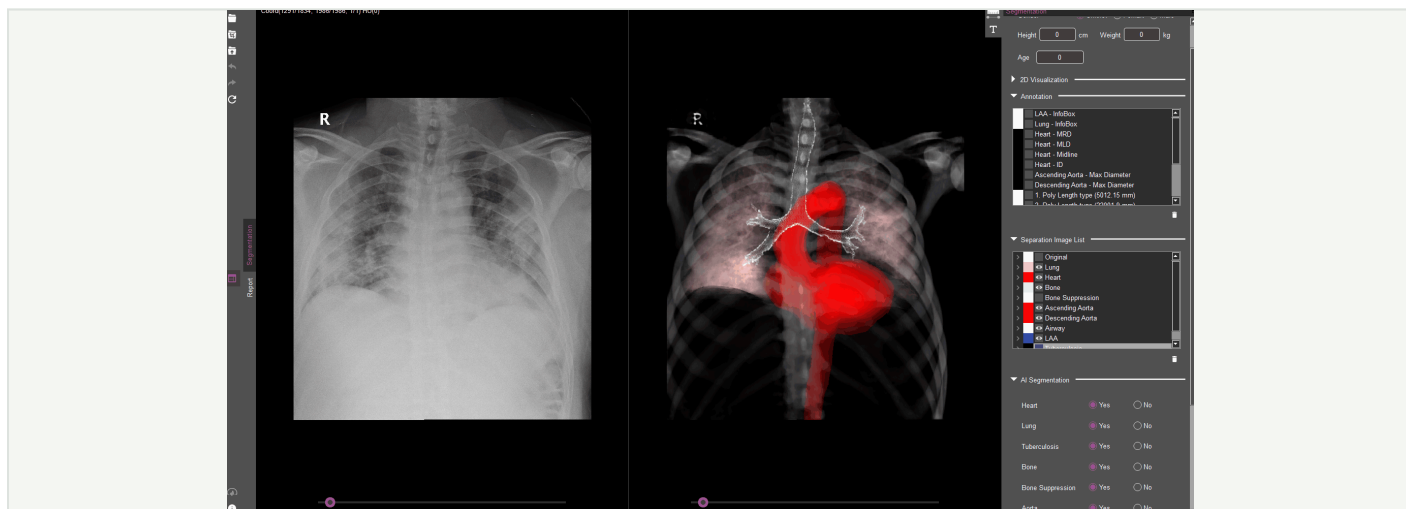
Composed AI results and generated images as report data and connected them to QWebEngine rendering and PDF output.

Analysis automation

Separated analysis and output work into Action and Command units executed sequentially through the Action Queue.

Institution-specific reports and packaging

Applied report templates, PDF/SVG resources, product options, package file lists, and upload conditions to institution-specific releases.



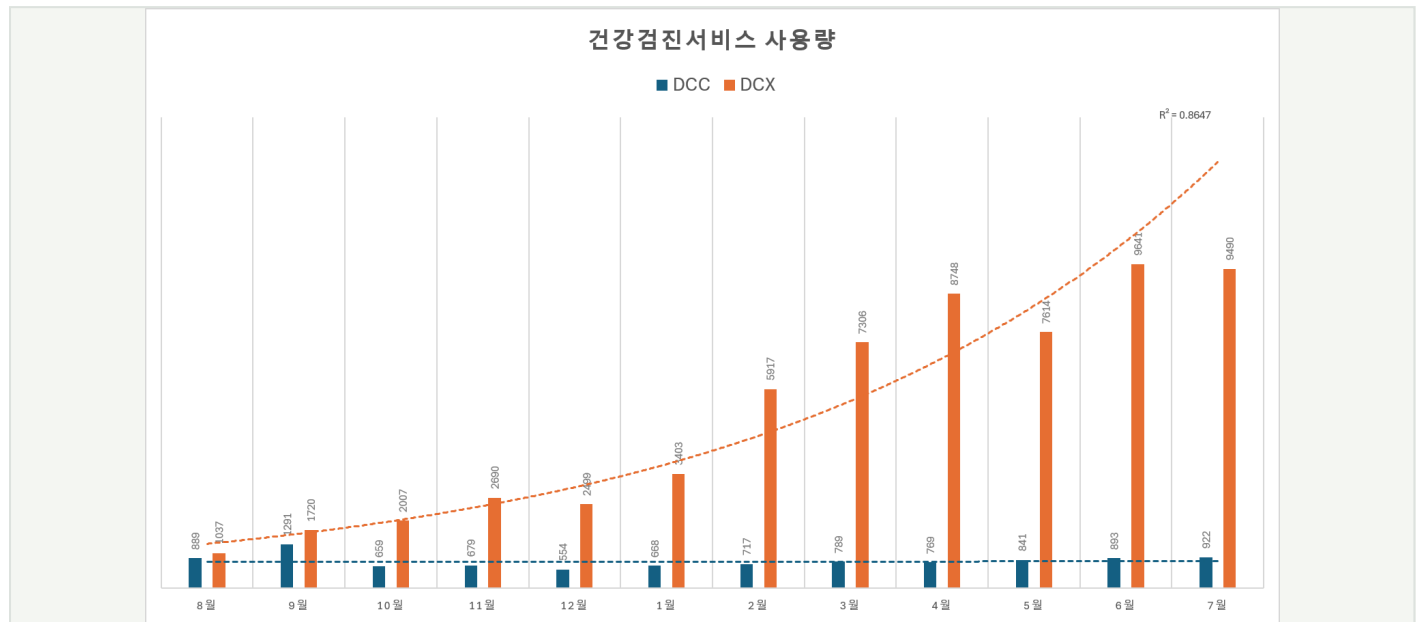
DeepCatch X client view · still from the original animation

C++ / Qt / MVVM / WW/WL / Segmentation / Annotation / QWebEngine / Action Queue / Product Option / Validation

Screening-site operations and usage

2025.08 - 2026.07

Product operations: DeepCatch services at 17 Korea Association of Health Promotion branches. DCX usage rose from 1,037 to 9,490, approximately 9.2 times, from August 2025 to July 2026 (company-reported figures).



Monthly DCX usage (Aug 2025-Jul 2026)

DCX usage: 1,037 in August 2025 → 9,490 in July 2026, approximately 9.2 times the starting value.



Korea Association of Health Promotion screening site

Screening and follow-up reports

Examples of institution-specific reports showing how chest X-ray results are presented as heart/aorta findings and cardiopulmonary health age assessments.



검사번호

1234567890

성명

홍길동

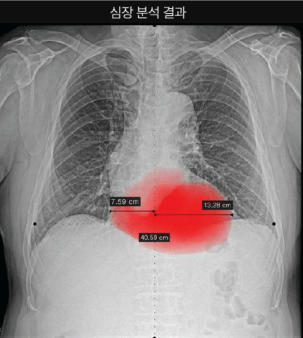
심장·대동맥 AI 분석 결과

검사 종류	정상 기준	검사 결과
상행대동맥 최대 직경	4.5 cm 미만	6.9 cm 요관리
하행대동맥 최대 직경	3.5 cm 미만	4.1 cm 요관리
가슴대비 심장비율 (심흉곽비)	50 % 미만	51 % 요관리

홍부대동맥 분석 결과



심장 분석 결과



X-ray 에서 찾은 홍부대동맥 크기

- 검사 결과가 상행대동맥은 4.5cm, 하행대동맥은 3.5cm 이상일 경우 대동맥류로 의심할 수 있습니다.
- 홍부대동맥류 관련 증상으로는 가슴과 등, 어깨, 허리 등에 통증입니다.

X-ray 에서 찾은 심장 크기

- 심흉곽비란 흉부에서 심장이 차지하는 비율로, 심장의 비대 여부를 확인하는데 사용되는 지표입니다.
- 심흉곽비가 50% 이상인 경우, 기저질환이 있거나 호흡곤란, 부종 등의 임상증상이 동반된다면 정밀검사를 통해 확인이 필요하며 증상이 없더라도 정기적인 추적검사를 권장합니다.

Heart and aorta report example



등록번호

1111111

성명

홍길동

성별/나이

M / 45

날짜

2025.08.29

AI 분석 심폐 건강나이

AI 분석 심폐 건강 나이는 흉부 X-ray를 AI로 분석해 심장·폐의 구조 변화를 평가하고, 같은 성별 연령대와 비교한 건강 나이입니다.

장기별 나이

심장 나이

46.4세

폐 나이

41.3세

나의 건강 나이는?

실제 나이

45.4세

저속 노화

-1.7세

건강 나이

43.7세

사망을 평균 00% 감소

나의 노화 속도는?

저속 노화

2006-2020년 간복상촬영원 건강검진 전체 수진자 중 동연령대 100명 중 상위 80% 노화 속도가 느린 편

심장 나이

당신의 심장 나이는 46.4세 입니다.

65등 / 100명

심장 평균 노화

심장 나이는 46.4세이며, 동일 성·연령대 대비 65 등 입니다.

심장 노화 속도는 실제 나이와 유사한 수준으로 진행되고 있습니다.

폐 나이

당신의 폐 나이는 41.3세 입니다.

5등 / 100명

폐 저속 노화

폐 나이는 41.3세이며, 동일 성·연령대 대비 5 등 입니다.

폐 노화 속도는 실제 나이보다 느린 수준으로 진행되고 있습니다.

종합 소견

심장은 실제 나이와 노화속도가 유사하며, 폐는 실제 나이에 비해 노화 속도가 느린 상태입니다. 현재 노화속도 유지 및 향상을 위해, 건강한 노화과정을 위해 정상체중 유지, 주 5회 이상 유산소 운동 (주 2회 이상 근력운동), 금연, 금주 또는 절주, 건강한 식습관, 충분한 수면 등 건강한 노화를 위한 생활습관을 유지하는 것이 도움이 될 수 있습니다.

Cardiopulmonary health age report example

CT reports from the related product family illustrate body composition follow-up and health management applications.

C++/Qt UI AutoTest Development

Aug 2023–Nov 2024 · First Project

Converted repeated manual UI checks into saved, replayable test scenarios.

Outcome Identified dozens of crashes through extended repeat testing and shared daily emails with test results and failure locations.

UI action recording

Recorded object names, event types, mouse positions, and input data as reusable test scenarios.

Input replay

Replayed buttons, combo boxes, sliders, line edits, trees, lists, tables, keyboard input, and shortcuts.

Long-running stability

Implemented repeat counts, popup handling, task completion waits, and step-by-step progress and result displays.

Validation improvement

Turned manual UI checks into reproducible flows with traceable failure points.

Scheduled runs and reporting

Connected scheduled TeamCity runs to daily emails containing test results and failure locations.

10	[AutoTest관련/DCX Research] [AutoTest-SUCCESSFUL]ITC-TESTPC03 DeepCatch X Research_beta_Install_v0.0.3.9641_with_AI.exe
10	[AutoTest관련/DCX Total] [AutoTest-FAILED]ITC-TESTPC03 DeepCatch X Total_Install_v1.0.0.9644_with_AI.exe
10	[AutoTest관련/DCX Aorta] [AutoTest-Stop]ITC-TESTPC03 DeepCatch X Aorta_Install_v1.1.0.9301_with_AI.exe
10	[AutoTest관련/DCX Total] [AutoTest-FAILED]ITC-TESTPC03 DeepCatch X Total_beta_Install_v1.0.0.9641_with_AI.exe
10	[AutoTest관련/MEDIP] [AutoTest-SUCCESSFUL]ITC-TESTPC02 MEDIP_beta_Install_v2.6.0.9641_with_AI.exe
10	[AutoTest관련/DeepCatch C] [AutoTest-FAILED]ITC-TESTPC02 DeepCatch_beta_Install_v1.3.1.9641_with_AI.exe
10	[AutoTest관련/DCX Research] [AutoTest-FAILED]ITC-TESTPC03 DeepCatch X Research_beta_Install_v0.0.3.9641_with_AI.exe
10	[AutoTest관련/MEDIP] [AutoTest-SUCCESSFUL]ITC-TESTPC02 MEDIP_beta_Install_v2.6.0.9641_with_AI.exe
10	[AutoTest관련/DeepCatch C] [AutoTest-SUCCESSFUL]ITC-TESTPC02 DeepCatch_beta_Install_v1.3.1.9641_with_AI.exe
10	[AutoTest관련/DCX Total] [AutoTest-FAILED]ITC-TESTPC03 DeepCatch X Total_beta_Install_v1.0.0.9641_with_AI.exe
10	[AutoTest관련/DCX Research] [AutoTest-FAILED]ITC-TESTPC03 DeepCatch X Research_beta_Install_v0.0.3.9641_with_AI.exe
10	[AutoTest관련/DCX Total] [AutoTest-FAILED]ITC-TESTPC03 DeepCatch X Total_beta_Install_v1.0.0.9639_with_AI.exe
10	[AutoTest관련/DCX Research] [AutoTest-FAILED]ITC-TESTPC03 DeepCatch X Research_beta_Install_v0.0.3.9639_with_AI.exe
10	[AutoTest관련/MEDIP] [AutoTest-SUCCESSFUL]ITC-TESTPC02 MEDIP_beta_Install_v2.6.0.9639_with_AI.exe
10	[AutoTest관련/DeepCatch C] [AutoTest-FAILED]ITC-TESTPC02 DeepCatch_beta_Install_v1.3.1.9639_with_AI.exe
10	[AutoTest관련/DCX Total] [AutoTest-FAILED]ITC-TESTPC03 DeepCatch X Total_Install_v1.0.0.9640_with_AI.exe
10	[AutoTest관련/DCX Total] [AutoTest-Stop]ITC-TESTPC03 DeepCatch X Total_Install_v1.0.0.9638_with_AI.exe
10	[AutoTest관련/MEDIP] [AutoTest-SUCCESSFUL]ITC-TESTPC02 MEDIP_beta_Install_v2.6.0.9639_with_AI.exe

AutoTest result email and failure reporting

C++ / Qt / UI Automation / TeamCity

MEDIP Batch Analysis and Export Macro

Jan-Apr 2026 · Batch Automation

Developed a long-running batch Macro that analyzes multiple DICOM and MIP inputs and saves multiple result formats.

Outcome Allowed processing to continue after individual file failures and recorded per-file status and error messages in JSONL to identify inputs requiring reprocessing.

Input and configuration

Accepted .dcm and .mip files or folders through drag and drop, recursively collected inputs, and saved target files, multiple AI projects, and export options as JSON.

Layer and mesh exports

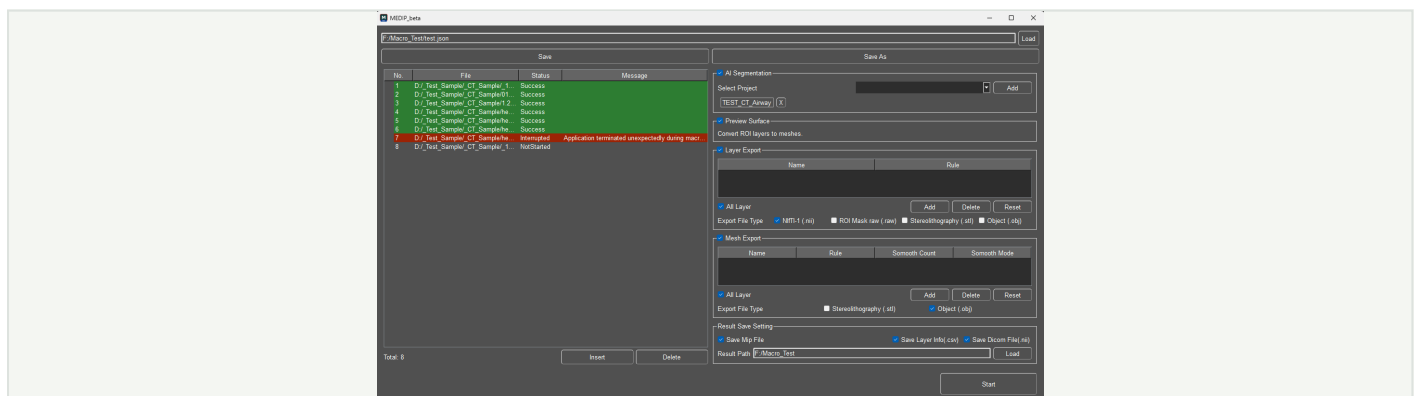
Selected layers by Start, End, or Contains name rules and exported NII, RAW, STL, and OBJ. Applied per-mesh Laplacian, Taubin, or HC Laplacian smoothing before STL/OBJ export.

Action Queue and failure isolation

Modeled each file as an independent Action and each processing stage as a Command. A failed file was recorded and removed while the queue continued with the next file.



Analysis and export options



Per-file execution status and failure messages

DICOM/PACS Stabilization and C-ECHO Addition

Jun 2024–Jun 2026 · Maintenance

Fixed search, download, and status-handling issues in an existing PACS integration and added asynchronous DICOM C-ECHO connectivity checks.

Outcome Made server connectivity visible in the product UI and resolved configuration differences and communication errors using reproducible conditions and logs.

Existing-feature maintenance

Analyzed and fixed operational issues in the existing Query/Retrieve, search, download, and DICOM transfer code rather than building those features from scratch.

DICOM C-ECHO

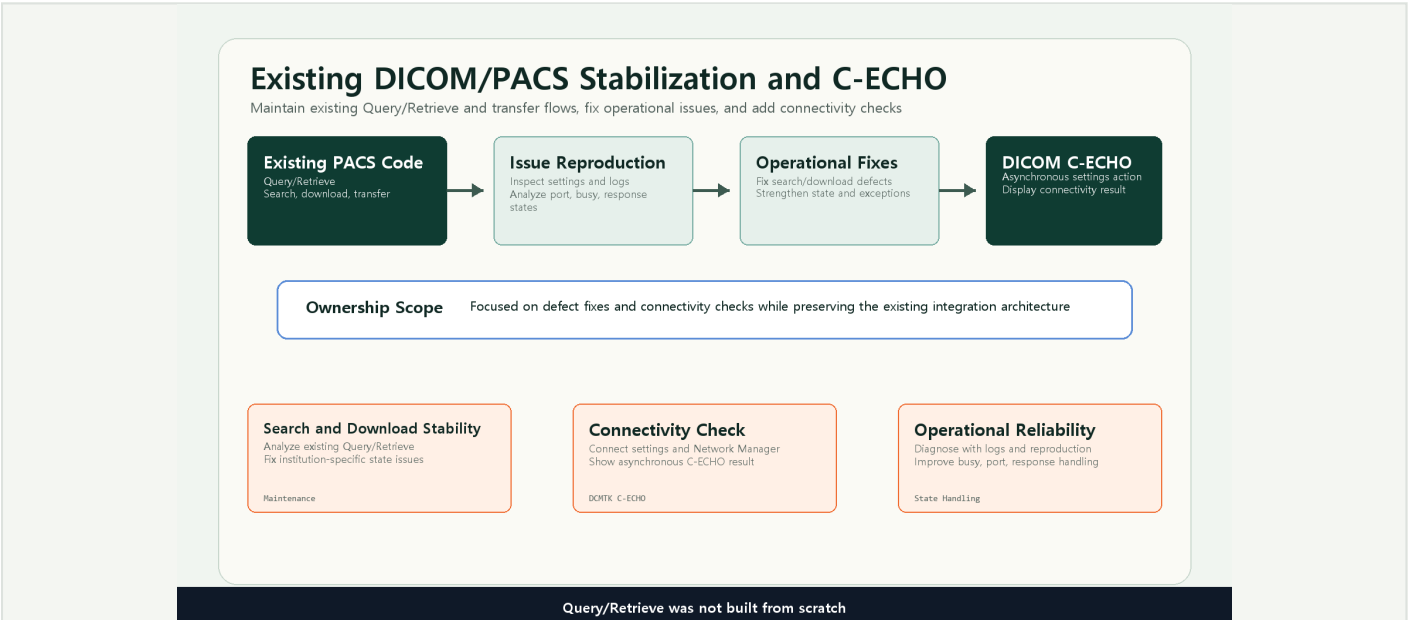
Connected an asynchronous Echo operation to PACS settings and the DICOM Network Manager for server connectivity checks.

Operational issue fixes

Used logs and reproduction conditions to resolve institution-specific settings, port state, and response-handling issues.

Clear ownership

Focused on stabilizing the existing integration and adding connectivity verification.



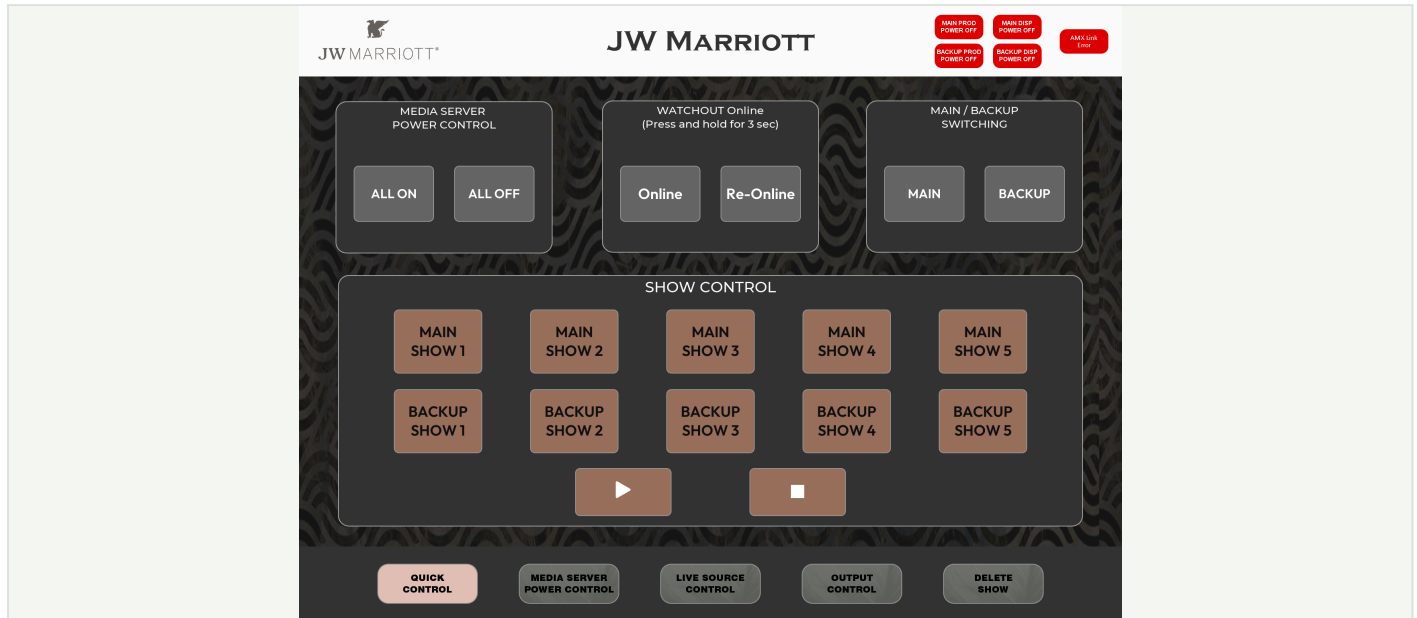
Existing integration stabilization and C-ECHO scope

C++ / Qt / DCMTK / DICOM C-ECHO / PACS

IMPEACH | R&D / Senior Staff

Oct 2019 - Feb 2023

Developed Python/Qt-based test applications and AMX-based integrated control programs. Implemented TCP/IP and Serial-based command/response flows for equipment and external software and handled WATCHOUT integration, connection states, and communication exceptions.



JW Marriott integrated control interface



THE ARC device control interface

Python / Qt / AMX / TCP/IP / Serial / WATCHOUT